

STUDENT SKILL DEVELOPMENT MANAGEMENT SYSTEM

The **Student Skill Development Management System (SSDMS)** is a web-based system developed to help students track and improve their skills in an organized way. In many educational institutions, information about a student's academic performance, technical skills, certifications, projects, internships and other activities is stored separately. Because of this, it can be difficult for students and faculty members to get a complete picture of a student's overall skill development. This system aims to solve that problem by bringing all the important skill-related information together in one place.

Using the system, students can create and manage their profiles, add their technical and soft skills, and mention their current and expected skill levels. They can also record assessment scores, certifications, projects and learning goals. One of the main features of the system is skill-gap analysis. It compares the student's current skill level with the target level and identifies the areas where improvement is needed. Based on the results, the system can suggest suitable learning activities, such as beginner or advanced courses, practice exercises, projects and certifications.

The system also helps students monitor their progress over time. A dashboard displays useful information such as the total number of skills, certifications and projects, along with the student's overall skill development progress. This makes it easier for students to understand their strengths and weaknesses and focus on areas that need more attention.

Faculty members and mentors can also benefit from the system by reviewing student skill profiles and providing feedback and guidance. Administrators can use the available information to understand common skill gaps among students and plan suitable training or development activities. This can support better academic planning and help students prepare for future employment opportunities.

The prototype was developed using **HTML, CSS and JavaScript**. HTML was used to create the structure of the web application, CSS was used to design a simple and responsive interface, and JavaScript was used to add interactive features such as adding skills, calculating skill gaps, tracking progress and generating learning recommendations. The system can also be extended using **XML, XSLT and JSP with the MVC architecture** to support structured data management and dynamic server-side processing.

In conclusion, the Student Skill Development Management System provides a simple and useful way to manage student skills and track their development. It helps students understand where they need to improve and supports faculty members in providing better guidance. In the future, the system can be improved by adding features such as AI-based recommendations, advanced analytics, mobile support and industry-based skill frameworks. Overall, the project demonstrates how web technologies can be used to create a practical solution for improving student skill development and employability.